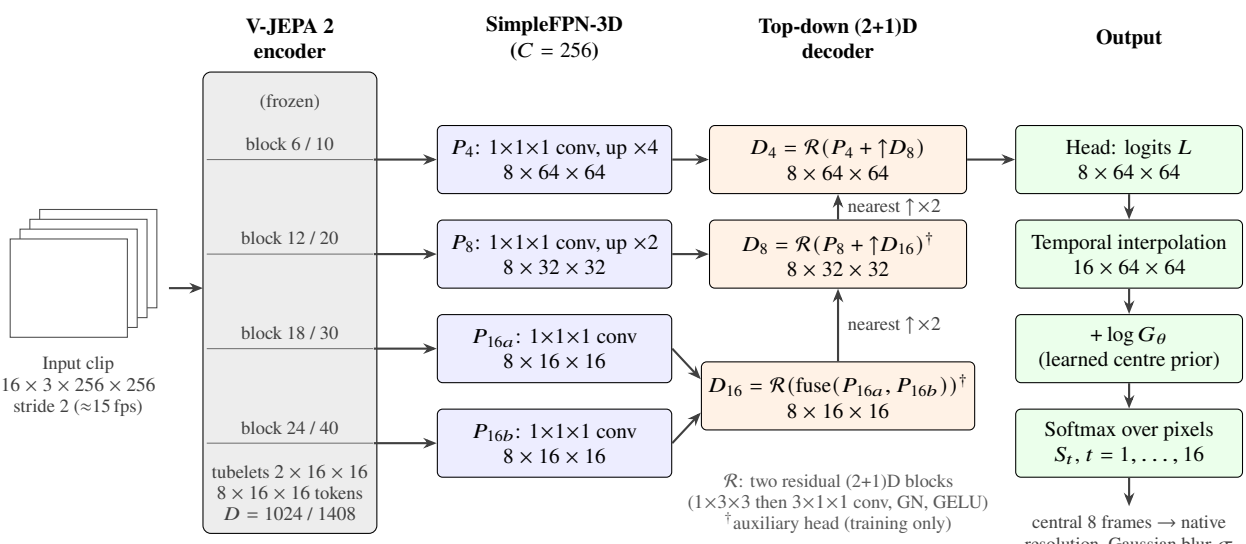

V-JEPA 2 encoder
SimpleFPN-3D ($C = 256$)
Top-down (2+1)D decoder
Output
(frozen)
block 6 / 10
block 12 / 20
block 18 / 30
block 24 / 40
tubelets $2 \times 16 \times 16$
$8 \times 16 \times 16$ tokens
$D = 1024 / 1408$
Input clip
$16 \times 3 \times 256 \times 256$
stride 2 (≈15 fps)
P_4: $1\times1\times1$ conv, up $\times4$
$8 \times 64 \times 64$
P_8: $1\times1\times1$ conv, up $\times2$
$8 \times 32 \times 32$
P_{16a}: $1\times1\times1$ conv
$8 \times 16 \times 16$
P_{16b}: $1\times1\times1$ conv
$8 \times 16 \times 16$
$D_4 = \mathcal{R}(P_4 + \uparrow D_8)$
$8 \times 64 \times 64$
nearest $\uparrow\times2$
$D_8 = \mathcal{R}(P_8 + \uparrow D_{16})^\dagger$
$8 \times 32 \times 32$
nearest $\uparrow\times2$
$D_{16} = \mathcal{R}(\text{fuse}(P_{16a}, P_{16b}))^\dagger$
$8 \times 16 \times 16$
$\mathcal{R}$: two residual (2+1)D blocks
($1\times3\times3$ then $3\times1\times1$ conv, GN, GELU)
†auxiliary head (training only)
Head: logits L
$8 \times 64 \times 64$
Temporal interpolation
$16 \times 64 \times 64$
$+ \log G_\theta$
(learned centre prior)
Softmax over pixels
S_t, $t = 1, \ldots, 16$
central 8 frames → native
resolution, Gaussian blur σ